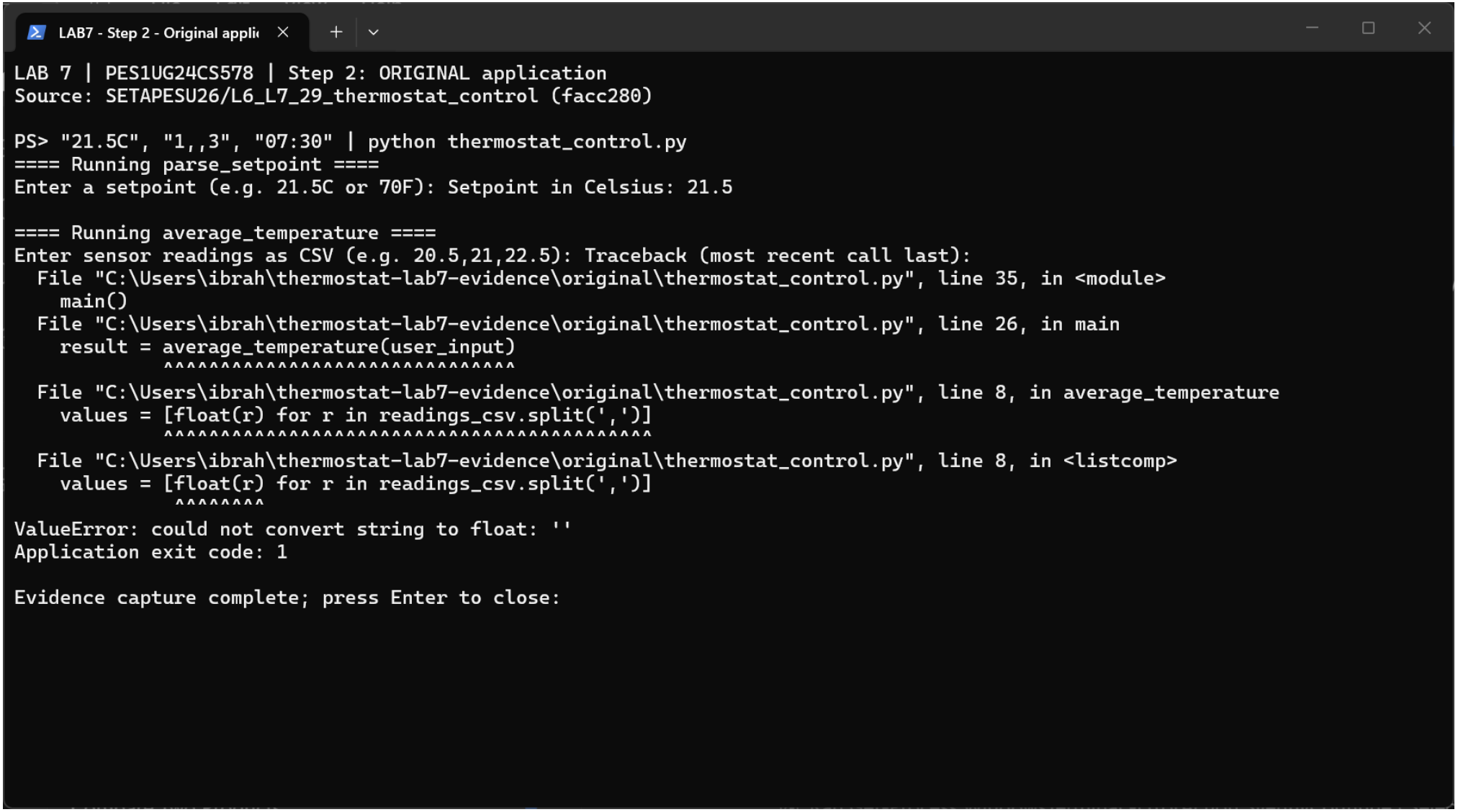
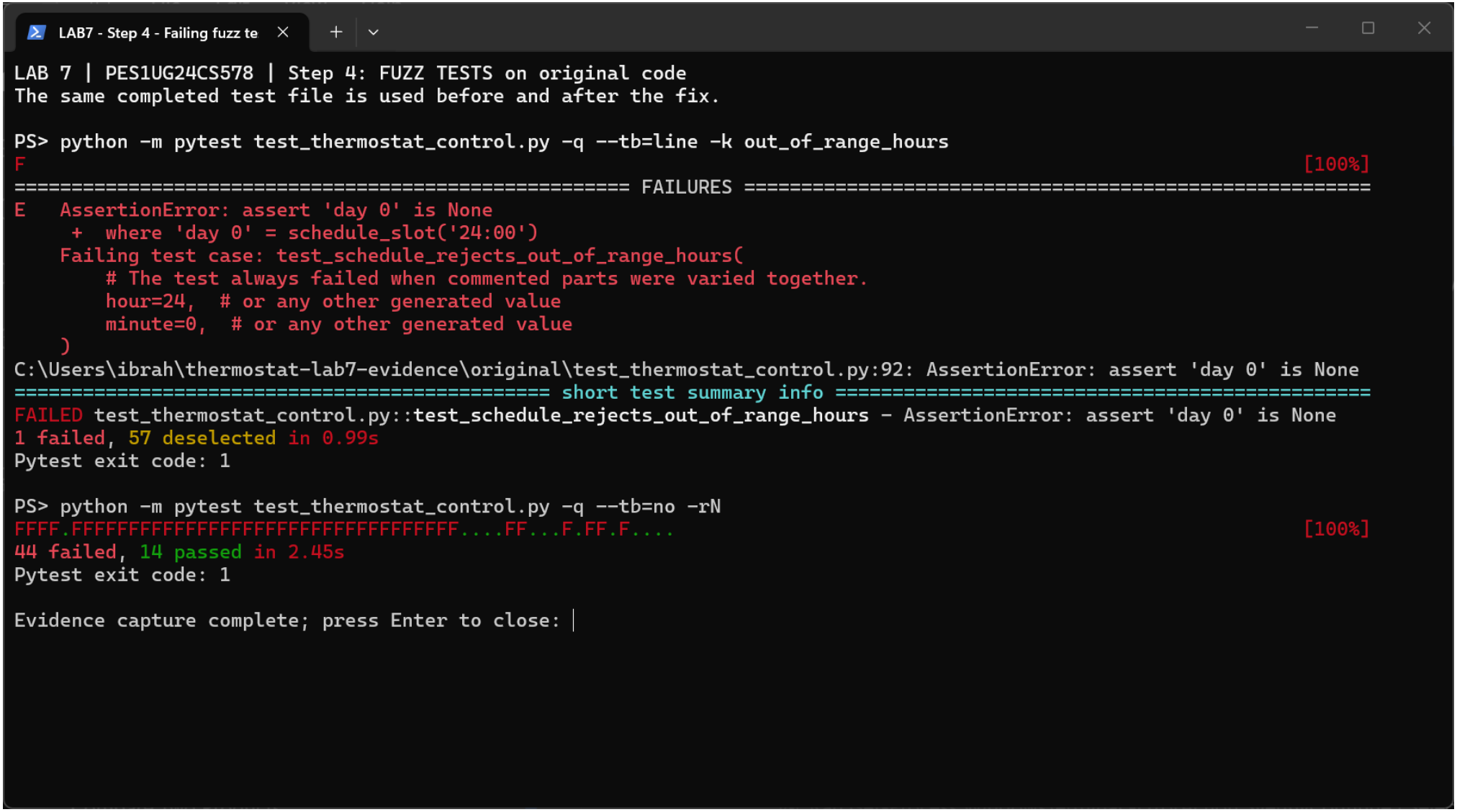
Step 2: Run the original application

Input 1,,3 contains an empty sensor reading. The original program raises ValueError and exits with code 1.



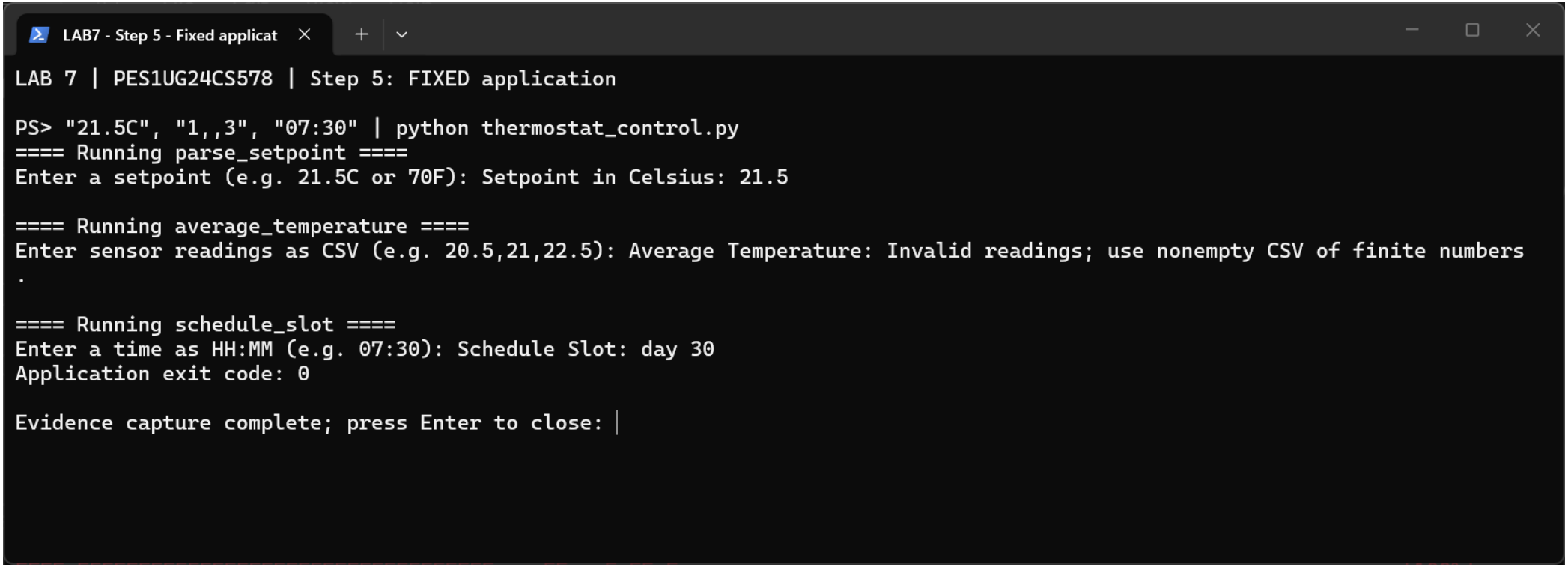
Step 4: Run the completed fuzz tests on original code

Hypothesis exposes the invalid time 24:00. The full unchanged suite records 44 failures and 14 passes.



Step 5: Run the fixed application

The same inputs now produce a validation message, continue to schedule_slot, and exit successfully.



Fixes applied

All three parsers return None for invalid types, empty values and malformed strings.

Setpoints require C/F units and finite values. CSV parsing rejects missing and non-finite readings.

A stable mean avoids overflow and cancellation. Times require ASCII HH:MM in 00:00-23:59.

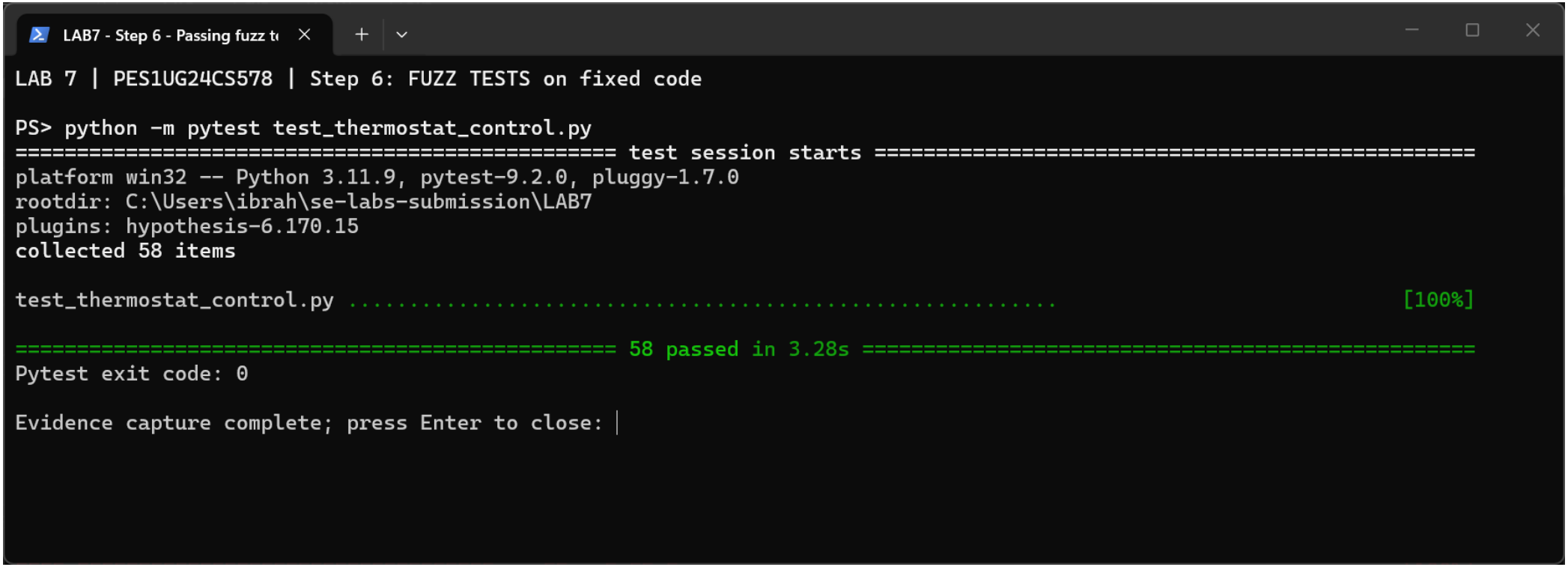
Surrounding whitespace is allowed; schedule labels use an unpadded minute for both day and night.

LAB 7: FUZZ TESTING FOR INPUT VALIDATION BUGS

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Step 6: Re-run the fuzz tests on fixed code

All 58 tests pass. The test file is identical in the before and after runs; pytest exits with code 0.



Test design and reproduction

8 Hypothesis properties (up to 500 examples each) plus 50 parametrized regression cases.

Strategies cover Unicode, None, wrong types, valid conversions, means, schedule boundaries and invalid ranges.

Windows / Python 3.11.9 / pytest 9.2.0 / Hypothesis 6.170.15. Captured on 11 November 2026.

Install: python -m pip install pytest hypothesis

Run inside LAB7: python -m pytest test_thermostat_control.py